Aya Alaoui Sossé

aya.alaouisossee@gmail.com • +212 650 594 635 • aya-alaoui-sossee.github.io • linkedin.com/in/aya-alaoui-sossé • github.com/aya-alaoui-sossee
Ph.D. Student in Computational Pathology | Data Scientist

Research Interests

Quantification of tumour-infiltrating lymphocytes (TILs) from H&E whole-slide images (WSIs) in triple-negative breast cancer (TNBC); measurement validity, confounder control, and cross-cohort generalization of digital pathology biomarkers for translational oncology and clinical research; and molecular prediction from histopathology (including EGFR mutation status in lung adenocarcinoma).

Education

Ph.D. in Artificial Intelligence & Data Science *2025 – Present*
Mohammed V University, Faculty of Science, Rabat, Morocco
Thesis: *AI-Driven Molecular Foundation Model for Cancer Pathology and Precision Medicine*. Focus: benchmarking and building a tissue- and reliability-aware TIL model in triple-negative breast cancer, validated against pathologist scoring and matched RNA-seq immune signatures before cross-cohort comparison and therapy response prediction.

M.Eng. in Big Data & Artificial Intelligence *2022 – 2025*
Euromed University of Fez (UEMF), Morocco. Full 100% merit scholarship for the 5-year engineering degree.
Coursework: Deep Learning, Medical Image Analysis, Statistical Modelling, Large-Scale Data Systems, Bioinformatics.

Summer Schools & Training

CCAIM Machine Learning for Healthcare Summer School, University of Cambridge *2026*
Awarded a funded place. Presented a 5-minute research talk and poster on validating AI-derived TIL measurements across patient cohorts in triple-negative breast cancer.

MAIR Mentorship Program Bootcamp, Moroccans in AI Research (MAIR) *May 2026*
Completed an intensive research training programme for early-stage AI researchers, delivered to a selected cohort.

HER2 AI Scoring Workshop, 2nd African Congress on Digital Pathology (AFCDP 2026), Agadir *April 2026*
Workshop delivered by Pr. Cécile Badoual, organised by the African Society of Digital Pathology (AFSDP).

Experience

Doctoral Researcher, Computational Pathology *2025 – Present*
Mohammed V University, Faculty of Science, Rabat, Morocco

- Benchmarking open-source TIL and pathology foundation models (WSInfer, ECTIL, MuTILs, H-optimus-0) in triple-negative breast cancer (TNBC) following TILs Working Group guidelines to identify failure mechanisms across tissue compartments and sites.
- Building a tissue- and reliability-aware TIL model validated across TCGA, public sTIL benchmarks (TIGER), and an independent clinical cohort (MetroHealth) against pathologist scoring and matched RNA-seq immune signatures (CD8A, GZMB/PRF1), prior to confounder-adjusted population comparison and therapy response prediction.
- Built an annotation platform for pathologists to annotate the private TIL cohort, including an AI-assisted read that pathologists can review, validate, or modify.

Data Scientist *2026 – Present*
DataPathology, Morocco

- Developed our first clinical MVP for H&E-based EGFR mutation prediction in lung cancer (foundation-model features and attention MIL on TCGA, CPTAC, and a Moroccan cohort), reaching a patient-level AUC of 0.74 and a zero-shot transfer AUC of 0.65 on Moroccan slides; detected and corrected a scanner-batch confounder and set up pre-registered, leakage-free evaluation.
- Work with digital pathology software companies to integrate their platforms into our labs for daily pathologist use, manage external partnerships, and helped 11 students get internships at DataPathology (3 built the company's first working demos).

R&D Intern, Medical AI *Feb – Aug 2025*
DataPathology, Morocco

- Built the first open-source whole-slide image pipeline at DataPathology for EGFR mutation prediction in lung adenocarcinoma, improving over baseline benchmark accuracy by 15%.

Undergraduate Research Projects *2023 – 2025*
Euromed University of Fez (UEMF), Morocco

- Built a multi-task deep learning model for joint spatial TIL quantification and adenocarcinoma vs. adenoma classification (2024–2025).
- Developed a breast cancer histopathology detection model reaching 94% accuracy against physician annotations and reducing slide review time by 60% (2023–2024).

Publications

Alaoui Sossé, A. et al. "A Systematic Mapping Study of Deep Learning for EGFR Mutation Prediction in Lung Adenocarcinoma." In: Idrissi, A. (ed.) *Modern Artificial Intelligence and Data Science 2026: Tools, Techniques and Systems (MAIDS 2026)*. Studies in Computational Intelligence, vol. 1311. Springer, Cham (Springer Nature Switzerland AG), ISBN 978-3-032-43857-7. Accepted / In Press.

Technical Skills & Languages

Whole-Slide Image Processing (OpenSlide, QuPath, TIAToolbox, TRIDENT), Pathology Foundation & TIL Models (H-optimus-0, RetCCL, ECTIL, WSInfer), Attention MIL, Batch-Confound Auditing, Leakage-Free Evaluation, Python, PyTorch, Hugging Face, Scikit-learn, Statistical Analysis (AUC, ICC, Spearman), Git, Linux.

Languages: Arabic (Native), French (C2), English (C2).

Awards, Conferences & Community

- **Speaker, AI-MEDX 2026 Global Summit on Artificial Intelligence in Medicine with eXplainability, UM6P** (Nov 2026) – Presented *"AI, Digital Pathology & Digital Twins: The Future of Lung Cancer Care"* (18 CPD hours).
- **2nd Place Finalist, GITEX Future Health Africa** (2026) – Pitched DataPathology on stage as one of three finalists and won 2nd place.
- **Industry Mentor, Durham University Computer Science Mentoring Programme** (2026 – Present) – Selected to mentor a Durham University Computer Science student on working in AI and technology.
- **Co-organiser, 1st and 2nd African Congress on Digital Pathology** (2025 – 2026) – Contributed to the scientific programme and organisation of both editions (2nd edition: AFCDP 2026, Agadir, organised by AFSDP); represented DataPathology at 2 international oncology AI conferences.
- **Volunteer English & Coding Instructor** (Mar 2023 – Feb 2025) – Taught English, basic algorithms, and logic to school students in summer camps.